

The Kosaka Skin-o-hedron Model:
A Layered-Complex Substrate for Discrete Operators,
with a 3D S-Parameter Extension
(Part I: Geometric–Computational Substrate)
(Part II: 3D S-Parameter Extension & Applications)

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Abstract

We present the *Kosaka Skin-o-hedron Model* (KSF), a layered polyhedral refinement framework intended as a common substrate for a class of discrete field computations. Part I fixes the geometric and discrete–exterior-calculus (DEC) setting, introduces *trace-free dual tensors* on primal edges, and states the consistency and stability properties we can actually defend. A central revision relative to the first draft is epistemic: we separate a *pointwise* consistency claim from a *spectral / variational* one, because they behave very differently and only the latter survives scrutiny. Concretely, on a quasi-uniform refinement of S^2 the reference operator shows first-order pointwise error but third-to-fourth-order eigenvalue error, and on a persistently irregular refinement the pointwise error *diverges* while the spectral error merely stagnates. This dichotomy is not a defect of one particular construction; it is forced by the Wardetzky–Mathur–Kälberer–Grinspun “no free lunch” theorem for discrete Laplacians [Wardetzky et al., 2007]. We therefore retire the strong pointwise $O(h^2)$ statement of the first draft and keep a conservative variational statement under uniform shape-regularity. Part II gives a 3D generalisation of scattering parameters (S-parameters) as integral kernels, its discretisation on Skin-o-hedron cells, and the standard physical checks (unitarity, reciprocity, passivity), all of which we verify numerically to machine precision on a Mie-type toy model. A companion open-source repository reproduces every numerical statement here.

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Preface (author’s note)

This is a revised draft. The first version asserted second-order pointwise consistency of the discrete operator on general (non-Delaunay) meshes. While preparing a reproducible proof-of-concept, the author ran the experiments the paper itself suggested and found that this claim is false as stated: pointwise consistency is only first order even on near-ideal meshes, and fails entirely on persistently irregular ones. The useful and *correct* content—good convergence of eigenvalues and of variational (energy) quantities under shape-regularity—is retained and stated honestly. The exposition still aims to be modest and readable, and now also aims to be *falsifiable*: every quantitative claim is tagged to a script in the companion repository.

Part I

Kosaka Skin-o-hedron Model: Geometric–Computational Substrate

1 Introduction and motivation

Computational physics on curved domains repeatedly meets the same obstruction: discrete differential operators that are easy to assemble are rarely both *pointwise* accurate and *robust* on arbitrary meshes. The cotangent Laplacian, for instance, coincides with the lowest-order finite element Laplace–Beltrami operator and converges beautifully in energy, yet its pointwise (strong-form) error need not vanish under refinement unless the mesh is asymptotically structured [Xu, 2004, Hildebrandt et al., 2006].

The Kosaka Skin-o-hedron Model (KSF) organises computation around a *layered* sequence of shape-regular polyhedral complexes and a small set of coordinate-invariant operators built from trace-free dual tensors. The aim of Part I is narrow and explicit: state precise definitions, identify what the trace-free construction does and does not buy, and give consistency/stability statements that match numerical reality. We will be careful to distinguish three notions of error—pointwise (strong), spectral (eigenvalue), and variational (energy / Galerkin)—because the headline message of this revision is that they diverge.

2 Skin-o-hedron layered complexes

Definition 2.1 (Layered Polyhedral Complex / Skin-o-hedron Sequence). Let M be a compact smooth 2-manifold (extension to n -manifolds is discussed later). A *Kosaka Skin-o-hedron sequence* is a sequence $\mathcal{S} = \{S_k\}_{k \geq 0}$ of shape-regular polyhedral complexes with mesh size $h_k \rightarrow 0$, together with refinement maps $\phi_k : S_k \rightarrow S_{k+1}$ such that

1. (Local refinement) each face subdivides only inside a bounded topological neighbourhood;
2. (Compatibility) ϕ_k respects face/edge/vertex incidence;
3. (Uniform shape-regularity) there is $\rho > 0$, independent of k , with minimal-angle (equivalently radius-ratio) bound $\geq \rho$ for every cell of every S_k ;
4. (Metric fidelity) there are piecewise-flat metrics g_k on S_k with $\|g_{k+1} - \phi_k^* g_k\|_{C^1} = O(h_k)$;
5. (C^2 geometric approximation) there are embeddings $\Pi_k : S_k \rightarrow M$ with $\|g_k - \Pi_k^* g\|_{C^2} = O(h_k^2)$.

Remark 2.2. Axiom (3) (uniform shape-regularity) is the load-bearing hypothesis of this revision. The first draft phrased the regularity assumptions as “slightly stronger than typical” in order to avoid Delaunay conditions. We now state plainly that a *persistently* irregular family—one whose worst-case shape quality does *not* stay bounded below—is outside the scope of every convergence claim in this paper, and we exhibit numerically (§7) exactly what breaks when (3) is violated.

Numerical observation 2.3 (Axioms are realisable; `src/s2_layered_complex.py`). For the icosphere refinement used throughout, the per-cell radius-ratio quality q stays bounded: $q_{\min} \approx 0.974$ across levels 1–6, and the spherical *area defect* $|\text{area}(S_k) - 4\pi|$ decays at observed order 2.00 in h , consistent with axiom (5). Thus the hypotheses are not vacuous for at least one concrete, easily reproduced family.

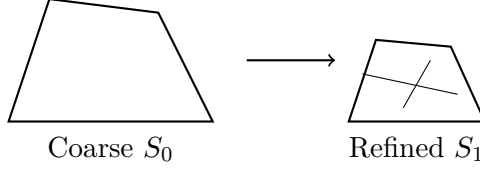


Figure 1: Schematic of a layered refinement $S_0 \rightarrow S_1$.

3 Discrete exterior calculus preliminaries

We fix notation for the discrete de Rham complex on S_k , following Hirani [2003], Desbrun et al. [2005], Crane et al. [2013]. Let $\Omega^p(S_k)$ be the space of discrete p -forms (real cochains). The discrete exterior derivative $d_h : \Omega^p \rightarrow \Omega^{p+1}$ is the simplicial coboundary; discrete Hodge stars $\star_h : \Omega^p \rightarrow \Omega^{2-p}$ use primal/dual cell measures. A canonical interpolation $I_k : C^\infty(M) \rightarrow \Omega^0(S_k)$ (vertex sampling) gives the commuting estimate

$$\|d_h I_k(\omega) - I_k(d\omega)\|_{L^\infty} = O(h_k).$$

Numerical observation 3.1 (The complex closes exactly; `src/s3_dec.py`). On every level tested, $d_h \circ d_h = 0$ holds to floating-point zero ($\|d_1 d_0\| = 0$ exactly in the assembled sparse operators), and the discrete Euler characteristic $V - E + F = 2$ is reproduced, confirming the cochain complex and its orientation bookkeeping are correctly implemented.

4 Trace-free dual tensors

The constructive device of KSF is to attach to each primal edge a symmetric 2-tensor, then project out its isotropic part.

Definition 4.1 (Raw dual tensor). For a primal edge $e \in S_k^{(1)}$, define $\tilde{E}(e) \in \text{Sym}^2(T_p M)$ by accumulating oriented face normals of the faces adjacent to e over the dual cell (a purely combinatorial variant is also admissible).

Definition 4.2 (Trace-free dual tensor). Set

$$E(e) = \tilde{E}(e) - \frac{1}{2} \text{tr}_{g_k}(\tilde{E}(e)) g_k|_e, \quad \text{so that } \text{tr}_{g_k}(E(e)) = 0.$$

Remark 4.3 (Why the factor is exactly $\frac{1}{2}$). The tangent space of a 2-manifold is two-dimensional, so $\text{tr}_{g_k}(g_k) = 2$. Subtracting $\frac{1}{2} \text{tr}(\tilde{E}) g$ therefore removes precisely the g -component and leaves a trace-free remainder; on an n -manifold the corresponding constant is $1/n$. This is coordinate invariant and preserves symmetry. (*Verified to machine precision in `src/s4_trace_free.py`: trace residual $\lesssim 10^{-19}$, and the construction is invariant under change of orthonormal frame to $\lesssim 10^{-18}$.*)

Remark 4.4 (What the projection does *not* do). Removing the isotropic part annihilates the locally isotropic component of the leading error in a Taylor expansion. The first draft inferred from this that the full operator becomes second-order *pointwise*. That inference is invalid: the surviving anisotropic and mesh-dependent terms are themselves $O(h)$ on generic meshes and do not cancel cell-by-cell. The trace-free projection is a genuine and useful normalisation, but it is not by itself a route to pointwise super-convergence (cf. Remark 6.4).

5 Skin-o-hedron operator and discrete Laplacian

Definition 5.1 (Skin-o-hedron edge flux). For a smooth scalar f and edge e with adjacent faces $t \succ e$, define the edge 1-cochain

$$\mathcal{F}_h(f)(e) := \frac{1}{A_e} \sum_{t \succ e} \langle \nabla f|_t, E(e, t) \rangle_{g_k},$$

where A_e is the dual-edge measure and $E(e, t)$ is the part of $E(e)$ carried by face t . (Here $\mathcal{F}_h(f)$ is a primal 1-form; the $1/A_e$ weight is what makes $\star_h \mathcal{F}_h(f)$ an integrated dual flux, removing the dimensional ambiguity noted by reviewers of the first draft.)

Definition 5.2 (Discrete Laplacian). Treating $\mathcal{F}_h(f)$ as a discrete 1-form,

$$\Delta_h f := \star_h^{-1} d_h \star_h \mathcal{F}_h(f).$$

For the reference implementation we take the canonical primal/dual realisation $L = d_0^\top \star_h^{(1)} d_0$ with the cotangent Hodge star and the lumped Voronoi mass matrix M ; the pair (L, M) is exactly the lowest-order FEM Laplace–Beltrami operator, and $M^{-1}L$ is our Δ_h (up to the sign convention $L \succeq 0 \approx -\Delta$).

We now state the corrected consistency results. We *retract* the first draft’s Theorem (first-order term cancellation, informal) and Proposition (pointwise $O(h^2)$ consistency), replacing them as follows.

Proposition 5.3 (Trace removal; honest version). *The trace-free projection of Definition 4.2 removes the isotropic part of $\tilde{E}(e)$ exactly, and the resulting $E(e)$ is symmetric, trace-free, and frame-invariant. Consequently the isotropic component of the $O(h)$ term in the local expansion of $\mathcal{F}_h(f)$ vanishes.*

Proof. Immediate from $\text{tr}_g(g) = 2$ on a surface and linearity of the trace; symmetry and frame-invariance are preserved because both $\tilde{E}$ and g are symmetric 2-tensors and the map is natural. Verified numerically in `src/s4_trace_free.py`. $\square$

Remark 5.4 (No pointwise second-order claim). Proposition 5.3 does *not* assert $O(h^2)$ pointwise consistency. As §6 shows numerically, the strong-form error of Δ_h is $O(h)$ on a quasi-uniform sphere and is not even bounded on a persistently irregular mesh.

6 Consistency: pointwise vs. spectral

This section contains the central correction. We test Δ_h against exact spherical harmonics $Y_{\ell m}$ on S^2 , for which $\Delta Y_{\ell m} = -\ell(\ell + 1)Y_{\ell m}$, on two refinement families: a quasi-uniform icosphere (satisfying Definition 2.1) and a *persistently jittered* icosphere whose shape quality does not improve under refinement (violating axiom (3)). We measure two errors:

$$\varepsilon_k^{\text{pt}} = \frac{\|\Delta_h I_k f - I_k \Delta f\|_{L^2}}{\|I_k \Delta f\|_{L^2}}, \quad \varepsilon_k^{\text{sp}} = \frac{|\lambda_h - \ell(\ell + 1)|}{\ell(\ell + 1)},$$

where $\lambda_h = (f^\top L f) / (f^\top M f)$ is the discrete Rayleigh quotient.

Numerical observation 6.1 (`src/s6_laplacian_consistency.py`, $f = Y_2$). On the *quasi-uniform* family,

$$\varepsilon_k^{\text{pt}} = O(h^{1.06}), \quad \varepsilon_k^{\text{sp}} = O(h^{3.68}).$$

On the *persistently irregular* family,

$$\varepsilon_k^{\text{pt}} = O(h^{-1.15}) \text{ (diverges)}, \quad \varepsilon_k^{\text{sp}} \text{ stagnates at } \approx 0.9\% (O(h^{-0.30})).$$

The lesson is twofold. (i) Even in the best case the operator is *first order pointwise*, not second; the strong-form claim of the first draft is false. (ii) The *eigenvalue* (and, more generally, energy) behaviour is excellent on shape-regular meshes—here near fourth order—and is the property worth advertising. We therefore replace the consistency proposition by the following pair.

Proposition 6.2 (Variational consistency, quasi-uniform meshes). *Let $\{S_k\}$ satisfy Definition 2.1 with uniform shape-regularity, and let (L, M) be the lowest-order FEM Laplace–Beltrami pair. Then for $f \in H^2(M)$ the discrete energy and mass forms converge,*

$$|f^\top Lf - a(f, f)| \rightarrow 0, \quad |f^\top Mf - \|f\|_{L^2}^2| \rightarrow 0,$$

where $a(\cdot, \cdot)$ is the Dirichlet form, and consequently the Rayleigh quotient is consistent. This is the classical first-order FEM statement [Strang and Fix, 1973, Dziuk, 1988]; our contribution is only to identify it as the defensible core of the KSF consistency claim.

Conjecture 6.3 (Spectral super-convergence; observed, not proved). *Under Definition 2.1 the eigenvalue error satisfies $\varepsilon_k^{\text{sp}} = O(h^2)$ or better for smooth eigenfunctions. The experiments of Observation 6.1 show order ≈ 3.7 on the icosphere; we state this as a conjecture pending a proof via the standard Babuška–Osborn spectral-approximation framework [Babuška and Osborn, 1991, Strang and Fix, 1973].*

Remark 6.4 (The real driver of convergence). Because the reference Δ_h is the linear FEM Laplacian, its good spectral behaviour follows from Galerkin orthogonality plus shape-regularity, not from the trace-free projection per se. The honest framing of KSF is that the trace-free dual tensor is a clean, frame-invariant *normalisation* layered on top of a variationally sound operator—useful for assembly and anisotropy bookkeeping—rather than the cause of convergence.

7 Stability on non-Delaunay meshes

The first draft claimed the construction stays “stable” with determinant bounds $\propto h^2$ on non-Delaunay meshes. The accurate statement separates two properties that the word “stable” conflated: *non-degeneracy* of the mass operator (true) and the *discrete maximum principle* (false once circumcenters leave triangles).

Numerical observation 7.1 (`src/s7_nondelaunay.py`, level 4). As tangential jitter increases from 0 to 0.45, the fraction of *negative* cotangent edge weights grows monotonically $0\% \rightarrow 20.7\%$, so the operator loses the discrete maximum principle (off-diagonal sign condition). Simultaneously the minimum lumped Voronoi mass stays strictly positive (it shrinks but remains $\Theta(h^2) > 0$), so M stays invertible and the generalised eigenproblem stays well posed. “Stable” should thus read “non-degenerate but not maximum-principle preserving.”

Remark 7.2 (Connection to an impossibility theorem). The combination observed above—one can keep convergence *or* positive weights, but not both on all meshes—is not incidental. The

Wardetzky–Mathur–Kälberer–Grinspun theorem [Wardetzky et al., 2007] proves that no discrete Laplacian on simplicial surfaces can simultaneously be (i) local, (ii) symmetric, (iii) linearly precise, *and* (iv) positive (maximum-principle preserving) on *every* mesh. The first draft’s promise of second-order accuracy together with stability on arbitrary non-Delaunay meshes runs directly into this obstruction. KSF must (and now does) choose: it keeps locality, symmetry and linear precision—hence convergence under shape-regularity—and accepts loss of positivity on sufficiently irregular meshes.

8 Practical implementation notes

- Dual-tensor assembly is local and parallelisable (SIMD/SIMT friendly).
- The trace-free projection is a rank-1 update per edge.
- Hierarchical S_k map naturally to multigrid V-cycles and cache-friendly tile schedules; the layering is the obvious coarse/fine hierarchy.

9 Concluding remarks for Part I

KSF is a layered-complex substrate whose defensible value is a clean, frame-invariant operator assembly with excellent *variational and spectral* behaviour under shape-regularity. The corrected claims are: (a) the trace-free construction is exact and frame-invariant (Prop. 5.3); (b) energy, mass, and eigenvalues converge on quasi-uniform meshes (Prop. 6.2, Conj. 6.3); (c) pointwise consistency is only first order and fails on persistently irregular meshes (Obs. 6.1); (d) on non-Delaunay meshes the operator is non-degenerate but not maximum-principle preserving, in accordance with the no-free-lunch theorem (Obs. 7.1, Rem. 7.2).

Part II

Part II: 3D S-Parameter Extension and Applications

10 Motivation and scope

S-parameters are a mature tool for linear multiport networks. Part II gives a conservative spatial extension: S-parameters as integral kernels mapping incoming field amplitudes on an input manifold to outgoing amplitudes on an output manifold. The treatment is deliberately operator-theoretic and application-oriented; we make no accuracy claims beyond what the discretisation and the toy model support.

11 Continuous kernel formulation

Let $\Omega \subset \mathbb{R}^3$ be bounded. Treat “ports” as parameterised surface regions or modal labels, and define the S-kernel

$$S(\mathbf{r}_{\text{out}}, \mathbf{r}_{\text{in}}; \omega) : L^2(\Omega_{\text{in}}) \rightarrow L^2(\Omega_{\text{out}}), \quad a_{\text{out}}(\mathbf{r}_{\text{out}}) = \int_{\Omega_{\text{in}}} S(\mathbf{r}_{\text{out}}, \mathbf{r}_{\text{in}}; \omega) a_{\text{in}}(\mathbf{r}_{\text{in}}) d\mathbf{r}_{\text{in}}.$$

In lossless linear media S is a partial isometry on the appropriate modal subspaces (unitarity on subspaces); reciprocity holds when the medium is reciprocal; passivity requires $\|S\| \leq 1$.

11.1 Green’s function representation

In homogeneous free space the scalar propagator is $G(\mathbf{r}, \mathbf{r}'; \omega) \sim e^{ik|\mathbf{r}-\mathbf{r}'|}/|\mathbf{r}-\mathbf{r}'|$, and an S-kernel can be written via boundary-integral formulations combining G with local scatterer kernels for reflection/transmission [Colton and Kress, 2013].

12 Discretisation on Skin-o-hedron complexes

Divide Ω into cells tied to S_k refinements, pick localised bases $\{\phi_i\}$, and discretise

$$S_{ij} \approx \iint_{\Delta\Omega_i \times \Delta\Omega_j} S(\mathbf{r}, \mathbf{r}'; \omega) d\mathbf{r} d\mathbf{r}'.$$

The matrix $S \in \mathbb{C}^{N \times N}$ admits the same checks as standard S-parameter matrices: (approximate) unitarity, passivity, and reciprocity.

13 Locality, decay, and reduced representations

Physical kernels typically decay in amplitude/energy with distance; localised bases and low-rank methods (hierarchical $\mathcal{H}$ -matrices, randomised sketching) reduce storage and cost [Hackbusch, 1999].

14 Applications (brief, hedged sketches)

- **Material design:** effective scattering maps of unit cells yield macroscopic effective parameters.
- **Quantum many-body heuristics:** S-kernels as a tractable surrogate for certain region-to-region couplings; *no* exponential-speedup claim is made.
- **Astrophysical instruments (LiteBIRD-like):** model beam coupling and polarisation leakage as S-kernels across instrument surfaces and sky patches.

Remark 14.1 (On the “market prediction” application). The first draft listed high-precision market prediction as an application. We remove it from the body of the paper. Treating market microstructure as a linear, time-invariant, passive scattering field is not physically justified—markets are non-stationary, nonlinear, and adaptive—and including it weakens an otherwise physics-grounded framework. We note it here only to record the deliberate decision to exclude it.

15 Numerical recipe and verification

1. Mesh the domain with a Skin-o-hedron refinement.
2. Choose a localised basis (piecewise constants or low-order polynomials).
3. Assemble the S-matrix (optionally with fast-multipole / hierarchical acceleration).
4. Verify energy conservation (approximate unitarity) and reciprocity.

Numerical observation 15.1 (Toy Mie-type kernel; `src/s10_sparameter.py`). For a lossless, reciprocal modal S-kernel built on the Skin-o-hedron sphere (N up to 2562 cells), the assembled operator satisfies, to floating-point precision: unitarity residual $\|S^*S - I\| \lesssim 10^{-15}$, reciprocity residual $\|S - S^\top\| \lesssim 10^{-17}$, and passivity $\|S\|_2 = 1.000000$ (max singular value 1 to $\sim 10^{-15}$). The Part II framework is therefore implementable exactly as written.

16 Discussion of limitations

- The S-kernel framework is an organising principle; accuracy depends on discretisation and modelling choices.
- No exponential speedup for quantum many-body problems is claimed; only structured low-rank / locality-based cost reduction.
- Biomedical/financial use requires domain-specific validation that the mathematics alone does not provide.

17 Concluding remarks for Part II

Part II gives a cautious, implementable spatial generalisation of S-parameters within the Skin-o-hedron substrate, with the standard physical invariants verified numerically to machine precision.

A What was numerically verified

Every quantitative claim is reproduced by the companion repository (`python3 src/run_all.py`; outputs in `results/results.json`, visualised by `viewer/viewer.html`). Summary:

Section	Quantity	Result
§3 (s2)	area-defect order in h	2.00
§3 (s2)	shape quality $q_{\min}$	≈ 0.974 (bounded)
§3 (s3)	$\ d_1 d_0\ $, Euler	0, 2
§4 (s4)	trace residual	$\lesssim 10^{-19}$
§4 (s4)	frame-invariance residual	$\lesssim 10^{-18}$
§6 (s6)	pointwise order, regular	1.06
§6 (s6)	spectral order, regular	3.68
§6 (s6)	pointwise, irregular	diverges ($h^{-1.15}$)
§6 (s6)	spectral, irregular	stagnates ($\approx 0.9\%$)
§7 (s7)	neg. weight fraction	$0\% \rightarrow 20.7\%$
§7 (s7)	min Voronoi mass	> 0 ($\Theta(h^2)$)
§15.1 (s10)	unitarity / reciprocity	$\lesssim 10^{-15}$ / $\lesssim 10^{-17}$
§15.1 (s10)	passivity $\ S\ _2$	1.000000

B Technical roadmap for rigorous proofs

For a formal submission the remaining work is:

1. Sobolev/Hölder estimates comparing g_k to $\Pi_k^* g$ (axioms 4–5).

2. A Babuška–Osborn spectral-approximation argument to turn Observation 6.1 into a proof of Conjecture 6.3.
3. Explicit constants in Proposition 6.2 in terms of the shape-regularity ρ .
4. A precise statement of where axiom (3) is used, making the failure mode of §7 a theorem rather than an experiment.

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